Optimal Four-to-Two-Bit Quantization Through a Binary Symmetric Channel

A finite computer-assisted proof with an equality classification

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Preprint notice. Not human peer-reviewed. Computer-assisted proof with reproducible finite certificates and reported independent AI reproduction. **Novelty: INCONCLUSIVE.** Full theorem-by-theorem comparison with T. Moazzeni (2026) remains incomplete; the claim of priority is expressly withheld. **License: All rights reserved.**

Abstract

Let X be uniform on the four-dimensional Boolean cube and let $Y = X \oplus Z$, where the coordinates of Z are independent $\mathrm{Ber}(p)$ variables. For every quantizer $Q : \{0,1\}^4 \to \{0,1\}^2$, we establish the bound

$$I(Q(X);Y) \le 2\big(1 - h_2(p)\big), \qquad 0 \le p \le \tfrac{1}{2}.$$

The bound is attained by selecting any two coordinates. For $0 < p < 1/2$, equality occurs exactly when the nonempty fibers of Q partition the cube into four two-dimensional faces: there are 30 unlabeled partitions and 720 maps with four fixed output labels. We prove a stronger inequality for uniform measures on arbitrary nonempty subsets of the cube. Its proof combines symmetry reduction, an exact-arithmetic interval certificate, a low-noise edge-boundary argument, and a high-noise Fourier estimate. This is a limited $n = 4$, $k = 2$ result, not a solution of the general multibit conjecture. The novelty of the special case relative to all prior results has not been established.

1 Problem and context

For $n, k \in \mathbb{N}$ and crossover probability $p \in [0, 1/2]$, consider the maximum of $I(Q(X);Y)$ over maps $Q : \{0,1\}^n \to \{0,1\}^k$, with X uniform and Y obtained by independently flipping each input coordinate with probability p. Projection onto k input coordinates yields $k(1-h_2(p))$ bits. For $k = 1$, the Courtade–Kumar problem was settled in recent work; the broader $2 \le k \le 7$ benchmark remains conjectural in Mahdavifar–Beirami [1]. An earlier result covers $k = n - 1$ [2].

Moazzeni's 2026 work [3] treats the fixed-k question and related high-noise/partition structure. Because the full text was not available for theorem-by-theorem comparison, the present special-case theorem is *not* claimed to be novel over that work. The purpose of this preprint is to present a specific, independently replayable finite proof candidate and its precise limitations.

Write

$$h_2(p) = -p\log_2 p - (1-p)\log_2(1-p), \quad h(p) = -p\ln p - (1-p)\ln(1-p), \quad \psi(p) = \ln 2 - h(p).$$

We use nats except when explicitly writing I or h_2 in bits; U_4 denotes uniform measure on $\{0,1\}^4$.

2 Main theorem and a stronger lemma

Theorem 1 (Four-to-two-bit quantization)**.** *Let $X \sim U_4$, $Y = X \oplus Z$ with Z_i independent* $\mathrm{Ber}(p)$*, and let $Q : \{0,1\}^4 \to \{0,1\}^2$ be arbitrary. For every $0 \le p \le 1/2$,*

$$I(Q(X);Y) \le 2\big(1 - h_2(p)\big) \quad \textit{bits.} \tag{1}$$

For $0 < p < 1/2$, equality holds if and only if Q has four nonempty fibers and each fiber is a two-dimensional face (subcube) of $\{0,1\}^4$. There are exactly 30 unordered four-face partitions and 720 quantizers with four labeled output symbols realizing equality. At $p = 0$ equality instead holds for every quantizer with four outputs of probability $1/4$; at $p = 1/2$ every quantizer is an equality case.

For any nonempty $A \subseteq \{0,1\}^4$, set $m = |A|$, let $X_A \sim \mathrm{Unif}(A)$ and let $q_{A,p}$ be the distribution of $X_A \oplus Z$.

Lemma 1 (Subset relative-entropy inequality)**.** *For all $A \neq \emptyset$ and $p \in [0, 1/2]$,*

$$D_{\mathrm{KL}}(q_{A,p} \| U_4) \le \log_2(16/m)\, \psi(p). \tag{2}$$

For $0 < p < 1/2$, equality holds exactly for subcubes A (including singletons and the entire cube).

Deduction of Theorem 1 from Lemma 1. Let $C = Q(X)$ and $A_c = Q^{-1}(c)$ for nonempty fibers. Uniformity of X and Y gives

$$I(C;Y) = \sum_c \frac{|A_c|}{16} D_{\mathrm{KL}}(q_{A_c,p} \| U_4) \le \psi(p) \sum_c \frac{|A_c|}{16} \log_2 \frac{16}{|A_c|} = \psi(p) H_2(C) \le 2\psi(p),$$

where $H_2(C)$ denotes entropy in bits; dividing by $\ln 2$ yields (1). For $0 < p < 1/2$, equality forces four equally likely outputs and equality in the subset lemma for each fiber, hence four faces. Conversely, each face achieves equality. Endpoint conditions follow from $Y = X$ when $p = 0$, and independence of X, Y when $p = 1/2$. □

3 Finite reduction and certificate design

The automorphism group of the four-dimensional cube, $(\mathbb{Z}_2)^4 \rtimes S_4$, has order 384. Enumerating its action on the 65,535 nonempty subsets yields 401 orbits. Exactly 81 subsets are subcubes, occupying five orbits; they satisfy (2) as an identity in p. Thus strict inequality requires checking only 396 orbit representatives. The accompanying verifier reconstructs the orbit decomposition rather than trusting a static list.

For each representative A, define

$$b_m = \log_2(16/m), \quad D_A(p) = D_{\mathrm{KL}}(q_{A,p} \| U_4), \quad B_m(p) = b_m \psi(p), \quad F_A(p) = B_m(p) - D_A(p).$$

The certificate divides the noise range into the three pieces below. Each assertion is checked in the accompanying standard-library Python implementation, with fail-closed assertions and SHA-256 manifests.

3.1 Central range: exact-rational interval covering

On $2^{-40} \le p \le 1023/2048$, both $D_A(p)$ and $B_m(p)$ are nonincreasing in p. The former follows from data processing for concatenated binary symmetric channels; the latter from monotonicity of binary entropy on $[0, 1/2]$. Therefore, for any rational interval $[l, r]$,

$$F_A(p) \ge B_m(r) - D_A(l) \quad \text{for all } p \in [l, r]. \tag{3}$$

The supplementary file `central_intervals.tsv` records 31,704 intervals, grouped by the 396 non-subcube orbit representatives. The verifier checks that every group's intervals cover exactly the central range without gaps or overlaps and recomputes each strictly positive integer-scaled lower bound in (3).

To enclose logarithms, normalize a positive rational as $x = 2^e t$, $1 \le t < 2$, and expand

$$\ln x = e \ln 2 + 2 \sum_{j=0}^{\infty} \frac{z^{2j+1}}{2j+1}, \qquad z = \frac{t-1}{t+1}, \quad 0 \le z < \tfrac{1}{3}.$$

Sixty terms, an explicit positive tail bound, and outward integer rounding at scale $S = 2^{128}$ yield certified enclosures for entropies and divergences. The smallest verified gap lower bound is approximately $1.772151923623782 \times 10^{-14}$ nats; the unit scale S^{-1} is approximately 2.94×10^{-39} nats. Positivity follows from inclusion-preserving outward bounds, not merely from comparing floating-point decimals.

3.2 Low-noise range: edge boundary

For $0 < p \le 2^{-40}$, let $E(A)$ be the number of cube edges crossing between A and its complement. The heat-flow entropy derivative admits an exact edge form,

$$\frac{dH(q_{A,p})}{dp} = \frac{1}{1-2p} \sum_{\{x,y\} \text{ edge}} (q_x - q_y)(\ln q_x - \ln q_y). \tag{4}$$

On a crossing edge, the inside probability is at least $(1-4p)/m$ and the outside probability at most p. Combining these facts with $\ln(1-4p) \ge -8p$ and $1/(1-2p) \ge 1$ yields the checked lower bound

$$F_A'(p) \ge \left[\frac{E(A)}{m}(1 - 5mp) - b_m \right] \ln(1/p) - \frac{E(A)}{m}(\ln m + 8p). \tag{5}$$

Here we explicitly use $h'(p) = \ln((1-p)/p) \le \ln(1/p)$ and $0 < 1 - 5mp \le 1$. The code verifies the strictly positive analytic lower bound for all 396 representatives uniformly on this range. Because $F_A(0) = 0$, integration gives $F_A(p) > 0$.

3.3 High-noise range: Fourier expansion

For $1023/2048 \le p < 1/2$, set $\rho = 1 - 2p \le 1/1024$. In Walsh–Fourier coordinates, write $16 q_{A,p}(y) = 1 + v(y)$ and $\widehat{v}(S) = \rho^{|S|} c_S$, with

$$c_S = \frac{1}{m} \sum_{x \in A} (-1)^{\sum_{i \in S} x_i}, \qquad L_1 = \sum_{|S|=1} c_S^2, \qquad L_{\text{hi}} = \sum_{|S| \ge 2} c_S^2.$$

With $\delta = (1+\rho)^4 - 1 < 1$, a Taylor bound and Parseval give

$$D_A(p) \le \frac{\rho^2}{2(1-\delta)} \left(L_1 + \rho^2 L_{\text{hi}} \right), \qquad \psi(p) \ge \frac{\rho^2}{2}.$$

It suffices to check

$$\frac{L_1 + 1024^{-2} L_{\mathrm{hi}}}{1 - [(1 + 1024^{-1})^4 - 1]} < b_m.$$

All 396 non-subcube representatives satisfy this strict bound, using exact rational Fourier coefficients and logarithm enclosures. At $p = 1/2$ both sides of (2) are zero.

4 Equality enumeration

There are $\binom{4}{2} 2^{4-2} = 24$ two-dimensional faces of $\{0,1\}^4$. The verifier independently enumerates all four-face selections that partition the 16 vertices, obtaining 30 unordered partitions. Labeling the four cells with the four output symbols gives $30 \times 4! = 720$ maps. Only the open interval $0 < p < 1/2$ admits this unique structural characterization. The two endpoints have the larger equality classes stated in Theorem 1.

5 Reproducibility and scope limitations

The accompanying source-and-evidence ZIP contains the finite certificate and Python 3.10+ standard-library verifier. Run `python -B verify_no49.py` from the evidence directory; it recomputes orbits, endpoint checks, exact quarter-noise bounds, interval coverage, each integer certificate margin, and face partitions. The checksum manifest detects changes to the frozen files. The source files were preserved byte-for-byte from the audited v0.2 evidence archive.

The result is finite computer-assisted mathematics, not a Lean/Coq formal proof. An independently implemented AI replication was reported by a separate reviewer, but its external code and full execution log are not bundled; no external human peer review has taken place. The claimed theorem applies only to $(n, k) = (4, 2)$ and does not settle the multibit conjecture for arbitrary n.

Novelty and priority note. We do not claim that the stated theorem is unpublished or independently original. A full comparison with the theorems and equality cases in Moazzeni [3] is outstanding. The present document should be read as a reproducible, provisionally positioned preprint. Subsequent overlap findings may require updating or withdrawing novelty claims.

AI and provenance statement

The manuscript, analytic reductions, finite certificate, and audit package were developed with AI assistance under the named author's direction. A second AI audit was reported separately. All responsibility for interpreting and publishing the result remains with the author. No external formal or human review is asserted.

References

[1] H. Mahdavifar and A. Beirami, "The Most Informative Bit and Beyond: A Proof of the Courtade–Kumar Conjecture and Multibit Extensions," arXiv:2609.26444 (2026). `https://arxiv.org/abs/2609.26444`.

[2] W. Huleihel and O. Ordentlich, “How to Quantize n Outputs of a Binary Symmetric Channel to $n-1$ Bits?,” *2017 IEEE International Symposium on Information Theory*, pp. 91–95. https://doi.org/10.1109/ISIT.2017.8006496.

[3] T. Moazzeni, “The Fixed-k Quantization Problem for Noisy Binary Sources,” *IEEE Transactions on Information Theory* (2026), DOI: 10.1109/TIT.2026.3741662. Bibliographic details and summary identified; full text not inspected as of this version.